

Innonsh SprintOS — Product Overview & Feature Guide

What is Innonsh SprintOS?

Innonsh SprintOS is a comprehensive internal product management and engineering execution platform. It acts as a single command center for software teams to plan sprints, assign tasks, track blockers, hold daily standups, collaborate in real-time, and measure team performance — all in one unified workspace.

It is built as a hybrid of Atlassian Jira, Azure DevOps Boards, and Agile Management Platforms, designed specifically for the internal workflows of Innonsh's engineering organization.

1. Key User Roles (Who Uses the Product?)

To keep data secure and maintain clear accountability, different team members have different access levels and capabilities.

Admin (System Owner)

Role: Has full administrative control over the entire platform. Key Tasks: Manages all users, projects, sprints, and global settings. Accesses security audit logs, login history, and session management. Can resolve blockers, archive/delete tasks, and override any workflow state.

Product Manager

Role: Oversees delivery, manages the product backlog, and drives sprint planning. Key Tasks: Creates and manages projects, plans sprints, creates/assigns tasks to developers, sets priorities, reviews team performance, resolves blockers, archives tasks, and views all analytics. Has the highest level of task control across all team members.

Developer (Team Member)

Role: Ground-level executor working directly with assigned tasks. Key Tasks: Updates task statuses on their assigned items, submits daily standup updates, logs blockers, adds comments, completes subtasks, and views their personal workload. Cannot access or modify other developers' private task assignments.

2. Core Business Modules & Features

Project Management (Project Hub)

What it does: Organizes all engineering work into projects, which serve as top-level containers for sprints and tasks.

Key Features:

Project Creation: Define projects with a name, description, unique key (e.g., INS-), status, and linked team members.

Project Lifecycle: Tracks project status across states — PLANNING, ACTIVE, ON_HOLD, and COMPLETED.

Project Actions: Product Managers can edit project details, change status, or delete projects as needed.

Project-linked Channels: Every project automatically gets a dedicated chat channel for all team communication related to that project.

Sprint Planning & Execution

What it does: Manages time-boxed delivery cycles (sprints) to organize tasks, set goals, and track progress toward a release.

Key Features:

Sprint Lifecycle: Each sprint moves through defined states — PLANNING → ACTIVE → COMPLETED — ensuring a controlled start and end.

Sprint Goals: Each sprint can have a named goal and defined start/end date, giving the team a shared objective.

Sprint Management: Product Managers can create, edit, and delete sprints with confirmation dialogs to prevent accidental changes.

Sprint-linked Channels: Active sprints get dedicated real-time chat rooms so developers can coordinate within the context of the sprint.

Sprint Health Panel: The dashboard displays a real-time view of the current sprint's health — completed vs. remaining tasks, blockers, and overall progress at a glance.

Task Management (Work Items)

What it does: The core unit of delivery. Every piece of work is tracked as a task linked to a project and sprint.

Key Features:

Task Creation: Tasks include title, description, acceptance criteria, priority, due date, story points, labels, assignee, and reporter.

Priority Levels: Five tiers — CRITICAL, URGENT, HIGH, MEDIUM, LOW — color-coded across the interface for instant visibility.

Status Workflow: Tasks progress through structured states: TODO → IN PROGRESS → IN REVIEW → TESTING → DONE. A BLOCKED state is available to flag impediments.

Subtasks: Break large tasks into smaller checklist items, each trackable as completed or pending.

Story Points: Estimate effort using story points to feed velocity calculations in Analytics.

Task Labels: Tag tasks with custom labels for additional categorization and filtering.

Attachments: Upload and view files directly within a task for context and documentation.

Archive & Restore: Product Managers can archive tasks to remove them from active board views without losing history, and restore them when needed.

Activity History: Every status change, comment, and update is logged in a per-task timeline for full auditability.

Standup Integration: Standup updates linked to a task are viewable directly inside the task detail drawer.

Agile Boards (Kanban View)

What it does: Provides a visual, drag-and-drop interface to manage task flow within a sprint.

Key Features:

Kanban Columns: Tasks are displayed as cards across status columns, providing an instant view of where work stands.

Drag & Drop: Developers and managers can drag task cards between columns to update status visually (powered by DND Kit).

Board Snapshot: The dashboard displays a mini board snapshot so managers can see sprint state without navigating away.

Blocker & Impediment Tracking

What it does: Flags tasks that are stuck and need escalation, ensuring blockers are visible and resolved quickly.

Key Features:

Blocker Logging: Developers can log a blocker against any task, specifying the blocker type and a description of what is preventing progress.

Blocker Visibility: Blocked tasks are visually highlighted in red across the board, task drawer, and dashboard.

Blocker Resolution: Product Managers and Admins can resolve blockers with a resolution note, which is recorded in the task's history.

Blocker Analytics: The Analytics Dashboard shows a breakdown of active blockers by category (e.g., Technical, External, Resource) as a donut chart.

Blocker Chat Rooms: Each blocker automatically spawns a dedicated chat discussion channel so the team can collaborate on resolution without cluttering other channels.

Daily Standups

What it does: Standardizes the daily team check-in, replacing informal message threads with a structured, logged format.

Key Features:

Standup Submission: Each team member submits a daily standup with three fields: Yesterday (what was done), Today (what is planned), and Blockers (any impediments).

Task Linking: Standup entries are linked to specific tasks, so each task's history shows which standups mentioned it.

Team Standup Monitoring: The dashboard provides a real-time view of who on the team has and has not yet submitted their standup for the day, giving managers instant accountability visibility.

Standup History: All past standup entries are stored and accessible per-developer and per-task.

Real-Time Collaboration Chat (SprintOS Chat)

What it does: An integrated Slack-like messaging platform built directly into SprintOS, eliminating the need for external chat tools for work-related discussions.

Key Features:

Channel Types: Organized into five distinct channel categories:

Announcements: Broadcast-only channels for company or team-wide updates.

Project Channels: Auto-created per project for project-wide team communication.

Sprint Channels: Auto-created per active sprint for focused sprint-level discussions.

Task & Blocker Discussions: Contextual chat rooms spawned directly from a task or blocker for focused resolution threads.

Direct Messages (DMs): One-on-one private conversations between any two team members.

Online Presence: Real-time user presence indicators (Online, Away, Offline) displayed on DM contacts.

Threaded Replies: Messages can be replied to in threads to keep focused sub-conversations organized.

Emoji Reactions: Users can react to messages with emojis.

@Mentions Autocomplete: Type @ to get an autocomplete dropdown of team members to notify them directly in a message.

Pinned Messages: Important messages in any channel can be pinned and viewed in a dedicated panel.

Message Editing & Deletion: Users can edit or delete their own messages.

Fuzzy Search: Search across all messages, channels, and discussions with a fast fuzzy search panel.

Unread Badge Counts: Each channel in the sidebar shows a live count of unread messages.

Announcement Banners: Pinned announcements can be displayed as persistent banners inside a channel.

Notifications System

What it does: Keeps every team member informed of relevant updates without requiring them to constantly check every module.

Key Features:

In-App Notifications: Real-time notifications for task assignments, status changes, mentions, standup reminders, and blocker alerts.

Notification Dropdown: Accessible from the top navigation bar with a live unread count badge.

Event-Driven: Notifications are triggered automatically by system events (task assigned, blocker raised, standup not submitted, etc.).

Search

What it does: Provides a global, cross-module search experience so users can find anything instantly.

Key Features:

Global Search: Search across projects, sprints, tasks, and team members from a single search bar.

Contextual Results: Results are categorized and displayed with relevant metadata for quick identification.

Reports

What it does: Generates formatted, exportable summaries of sprint and team performance for review or sharing.

Key Features:

Sprint Reports: Summarizes what was planned vs. completed in each sprint.

Retrospective Comparisons: Compares performance metrics across multiple sprints to identify trends over time.

Historical Data: Archived sprint and task data is retained for long-term reporting.

Analytics Dashboard

What it does: Provides a visual, data-driven view of how the engineering team is performing across projects, sprints, and individuals.

Key Features:

KPI Scorecards: At-a-glance metrics including:

Total Active Tasks

Sprint Completion Rate (%)

Delayed Tasks Count

Active Blockers Count

Team Velocity (story points)

Average Task Completion Time (hours)

Active Projects Count

Team Utilization (%)

Sprint Velocity Chart: A line chart comparing planned vs. completed story points across all historical sprints, showing the team's delivery trend.

Sprint Burndown Chart: An area chart tracking how remaining work burns down day-by-day within the active sprint vs. the ideal progress line.

Team Workload Distribution: A bar chart showing how many pending tasks are assigned to each team member, helping managers spot overloaded or underutilized developers.

Blockers by Category: A donut pie chart categorizing all active blockers (e.g., Technical, Resource, External) for quick prioritization.

Productivity Timeline: A time-series area chart showing the number of tasks completed and standups submitted per day, revealing productivity patterns over time.

Sprint Filter: All charts can be filtered by a specific sprint or viewed across all historical data.

Dashboard (Command Center)

What it does: The first screen after login — a real-time command center that surfaces the most critical information for both managers and developers.

Key Features:

KPI Cards: Key metrics from the active sprint displayed at the top.

Sprint Board Snapshot: A mini Kanban view of the active sprint's task distribution.

Sprint Health Panel: Overview of the current sprint's progress, blocked tasks, and at-risk items.

Team Standup Monitoring: Live list of team members who have and haven't submitted today's standup.

Team Workload Panel: Visual bar indicators showing each developer's current task load.

Blocked Task Cards: Highlighted cards for any currently blocked tasks requiring immediate attention.

Recent Standup Updates: A live feed of the latest standup submissions from the team.

Team & Member Management

What it does: Manages the people inside SprintOS — their roles, departments, and assignments.

Key Features:

Role-Based Access Control (RBAC): Each user is assigned a role (ADMIN, PRODUCT_MANAGER, DEVELOPER) that controls what they can see and do.

Department Grouping: Users belong to departments for organizational clarity.

Team Member Directory: Accessible from the Chat for starting direct messages with any colleague.

Avatar & Profile: Each user has a name, email, avatar, and role visible throughout the platform.

Security & Session Management

What it does: Protects the platform with enterprise-grade authentication and activity logging.

Key Features:

JWT Authentication: Access tokens (15-minute expiry) and refresh tokens (7-day expiry) stored as secure HTTP-only cookies.

Refresh Token Rotation: Every token refresh rotates the refresh token, detecting and blocking stolen token reuse.

Brute-Force Protection: Locks accounts after 15 failed login attempts within 15 minutes and logs suspicious activity.

New IP / New Device Detection: Flags logins from new IP addresses or unrecognized devices with a MEDIUM severity security alert.

Login History: Full log of every login attempt (success, failure, locked) including IP address, device, and browser.

Security Audit Log: Persistent log of all security-relevant events (LOGIN_SUCCESS, LOGOUT, SUSPICIOUS_NEW_IP, SUSPICIOUS_BRUTE_FORCE) for compliance and review.

Session Revocation: Sessions can be invalidated on logout or on detection of security anomalies.

3. Technology Platform

Layer	Technology
Frontend	React, TypeScript, Vite, TailwindCSS, shadcn/ui
State Management	Zustand, React Query
Real-Time	Socket.IO
Backend	Node.js, Express.js, TypeScript
ORM	Prisma ORM
Database	PostgreSQL (Supabase)
Auth	JWT (Access + Refresh tokens), bcrypt
Deployment	Frontend → Vercel, Backend → Render
Infrastructure	Docker, GitHub Actions CI/CD